

**VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY
INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**



**WEB APPLICATIONS DEVELOPMENT
IT093IU
LAB 7**

Member List:

- Huỳnh Ngọc Bảo Long – ITITWE23023

Instructor: MSc. Nguyễn Trung Nghĩa

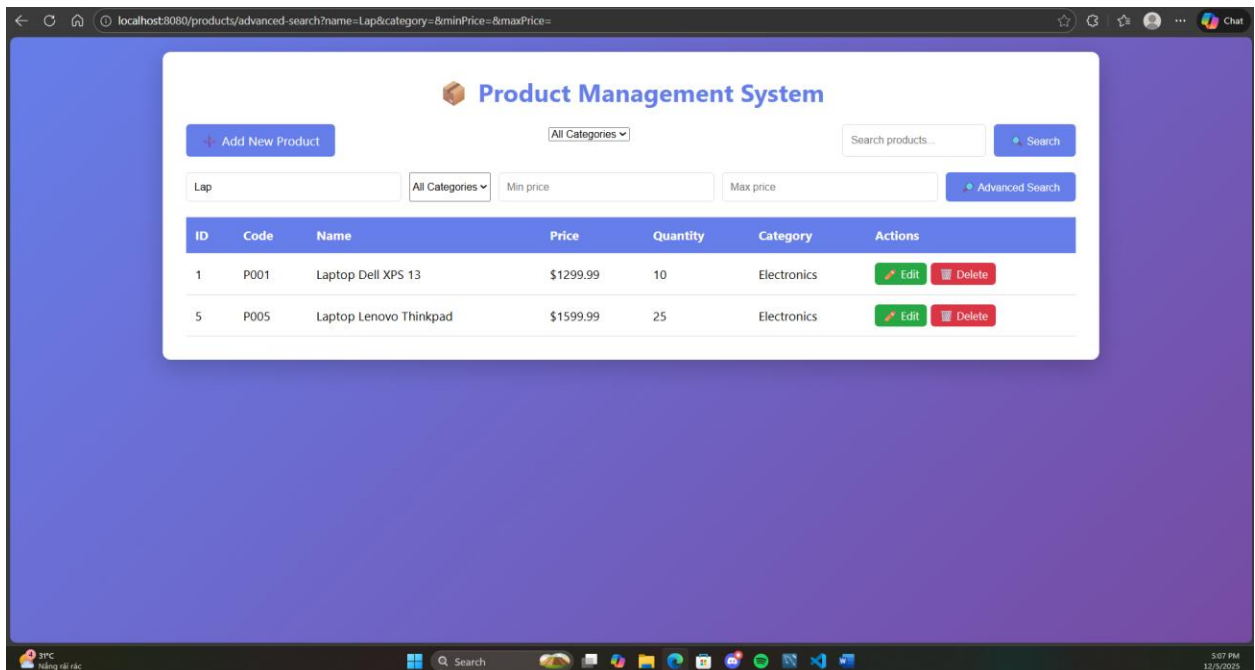
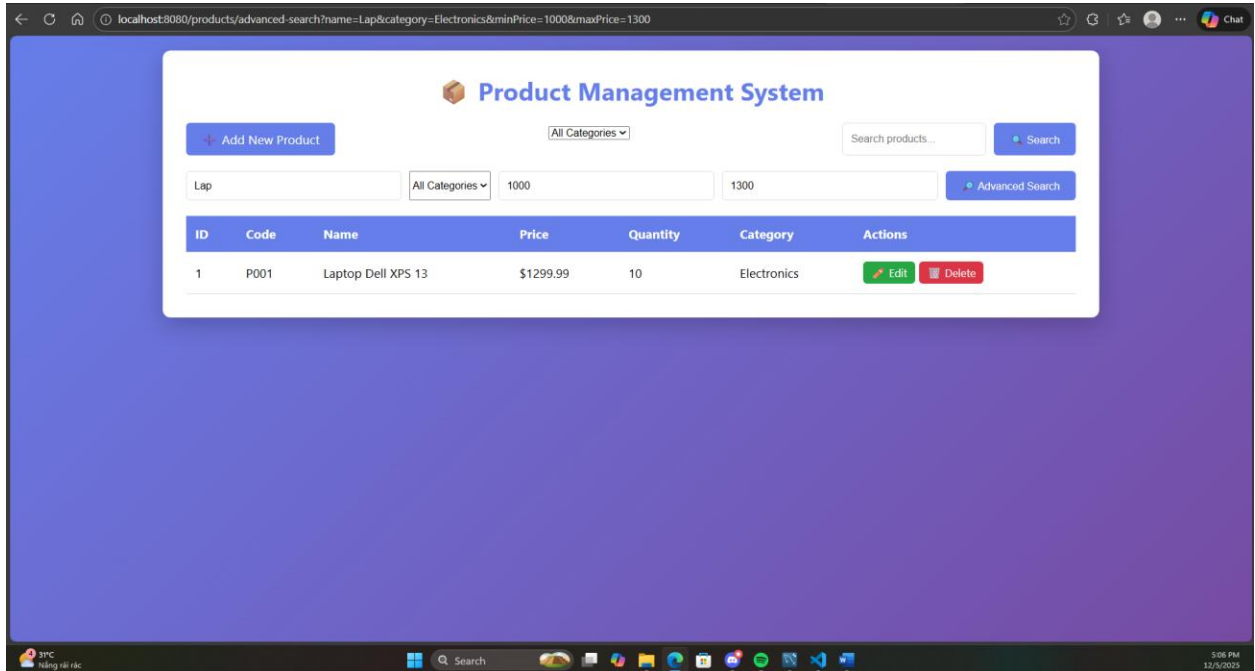
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Exercise 5

1. Demo/Test case

Search by criteria (Advanced search)



localhost:8080/products/advanced-search?name=&category=Books&minPrice=&maxPrice=

Product Management System

[+ Add New Product](#) All Categories ▾ [Search](#)

All Categories ▾ [Advanced Search](#)

ID	Code	Name	Price	Quantity	Category	Actions
10	P010	Principles of Database Management	\$4.99	10	Books	Edit Delete

31°C Hàng rất nice 5:09 PM 12/4/2023

localhost:8080/products/advanced-search?name=&category=&minPrice=0&maxPrice=100

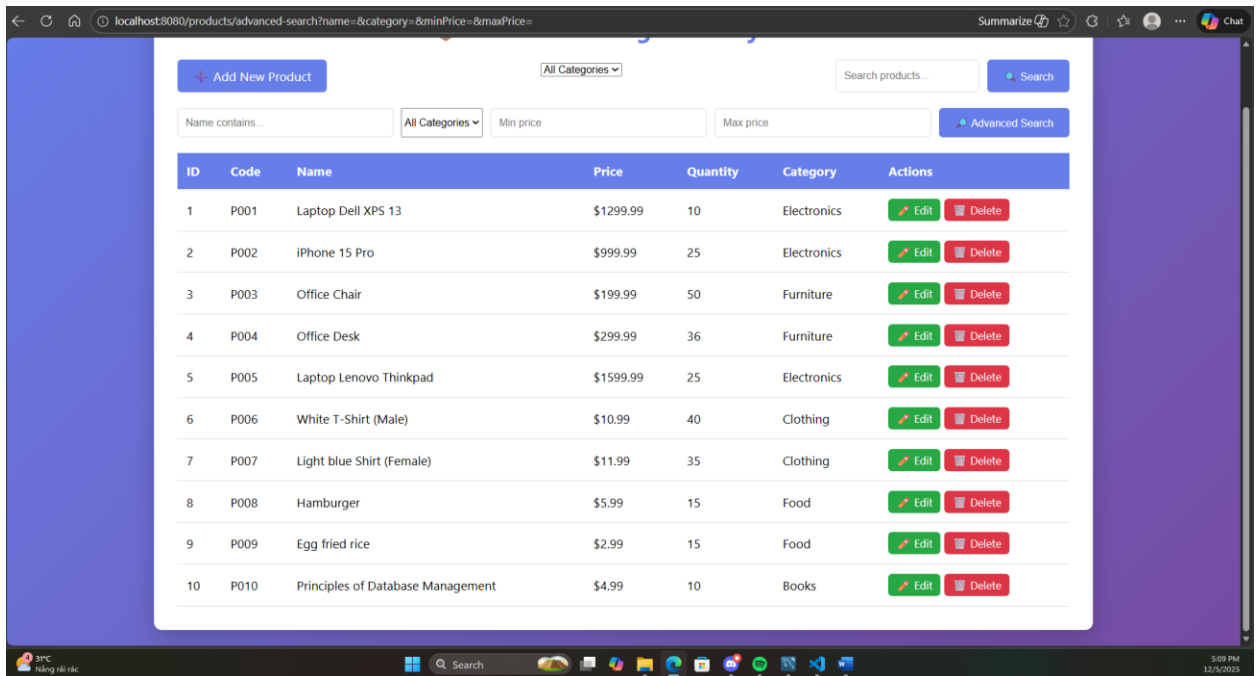
Product Management System

[+ Add New Product](#) All Categories ▾ [Search](#)

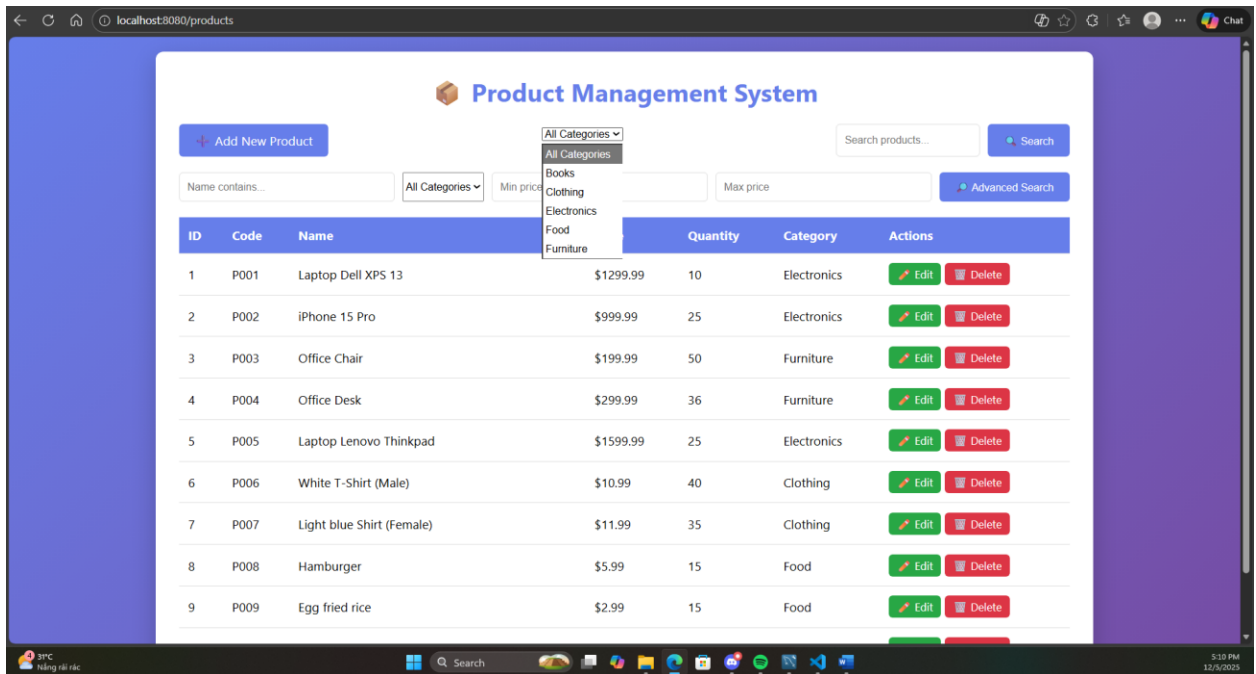
All Categories ▾ [Advanced Search](#)

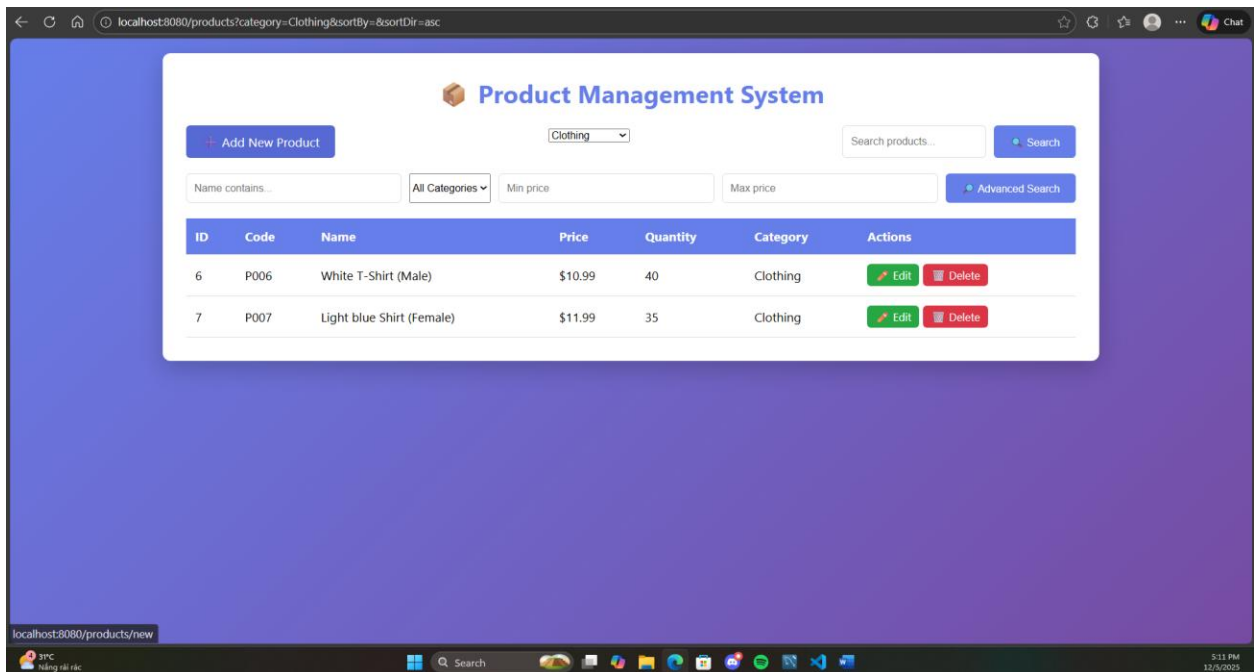
ID	Code	Name	Price	Quantity	Category	Actions
6	P006	White T-Shirt (Male)	\$10.99	40	Clothing	Edit Delete
7	P007	Light blue Shirt (Female)	\$11.99	35	Clothing	Edit Delete
8	P008	Hamburger	\$5.99	15	Food	Edit Delete
9	P009	Egg fried rice	\$2.99	15	Food	Edit Delete
10	P010	Principles of Database Management	\$4.99	10	Books	Edit Delete

31°C Hàng rất nice 5:09 PM 12/4/2023

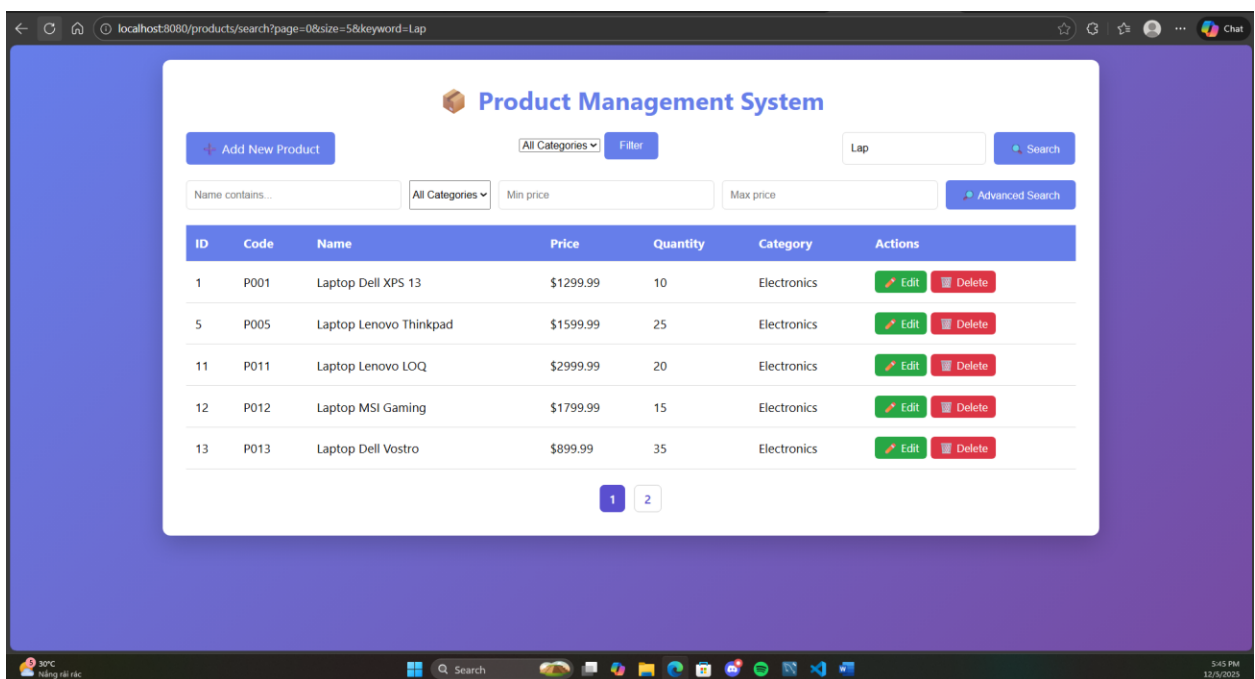


Category filter





Pagination



2. Code explanation

- Repository

- The @Query annotation allows optional parameters: (:name IS NULL OR p.name LIKE %:name%)
- This ensures every field is optional.

- Service
 - Service forwards values to repository for execution.
- Controller
 - Reads query parameters and passes them to the service.
- View
 - Thymeleaf form sends keyword, category, price min/max.

Exercise 6

1. Demo/Test case

The screenshot shows a web browser window with the URL `localhost:8080/products/save`. The page displays a form titled "+ Add New Product". The form contains the following fields and values:

- Product Code ***: A text input field with the placeholder "Enter product code (e.g., P001)". Below the field, three red error messages are displayed: "Product code must be 3-20 characters", "Product code is required", and "Product code must start with P followed by numbers".
- Product Name ***: A text input field containing the value "Algorithms and Data Structures".
- Price (\$) ***: A text input field containing the value "14.99".
- Quantity ***: A text input field containing the value "35".
- Category ***: A dropdown menu with "Books" selected.
- Description**: A text area containing the value "Book for Algorithms and Data Structures".

The browser's taskbar at the bottom shows the system clock as 5:50 PM on 12/9/2025.

Add New Product

Product Code *

AA23

Product code must start with P followed by numbers

Product Name *

Algorithms and Data Structures

Price (\$) *

14.99

Quantity *

35

Category *

Books



Description

Book for Algorithms and Data Structures



Save Product



Cancel

Add New Product

Product Code *

P016

Product Name *

ab

Name must be 3-100 characters

Price (\$) *

14.99

Quantity *

35

Category *

Select category



Category is required

Description

Enter product description (optional)



Save Product



Cancel

Add New Product

Product Code *

P016

Product Name *

Algorithms and Data Structures

Price (\$) *

-5

Price must be greater than 0

Quantity *

-1

Quantity cannot be negative

Category *

Books



Description

Enter product description (optional)



Save Product



Cancel

2. Code explanation

- Entity-level validation
 - Annotations like:

- @NotBlank
- @Pattern
- @DecimalMin

tell Spring Boot to validate fields automatically.

- Controller

- @Valid triggers validation; BindingResult captures errors.
- If errors exist: return "product-form";
so page displays errors.

- View

- Thymeleaf shows field errors: `<span th:errors="*{name}"></span>`

Exercise 7

1. Demo/Test case

Name ↑↓
Price ↑↓

[Add New Product](#)

All Categories

Filter

Search

All Categories

Min price

Max price

Advanced Search

ID	Code	Name	Price	Quantity	Category	Actions
6	P006	White T-Shirt (Male)	\$10.99	40	Clothing	Edit Delete
10	P010	Principles of Database Management	\$4.99	10	Books	Edit Delete
4	P004	Office Desk	\$299.99	36	Furniture	Edit Delete
3	P003	Office Chair	\$199.99	50	Furniture	Edit Delete
7	P007	Light blue Shirt (Female)	\$11.99	35	Clothing	Edit Delete
12	P012	Laptop MSI Gaming	\$1799.99	15	Electronics	Edit Delete
5	P005	Laptop Lenovo Thinkpad	\$1599.99	25	Electronics	Edit Delete
11	P011	Laptop Lenovo LOQ	\$2999.99	20	Electronics	Edit Delete
1	P001	Laptop Dell XPS 13	\$1299.99	10	Electronics	Edit Delete

Product Management System

[Add New Product](#) All Categories Filter Search

All Categories Advanced Search

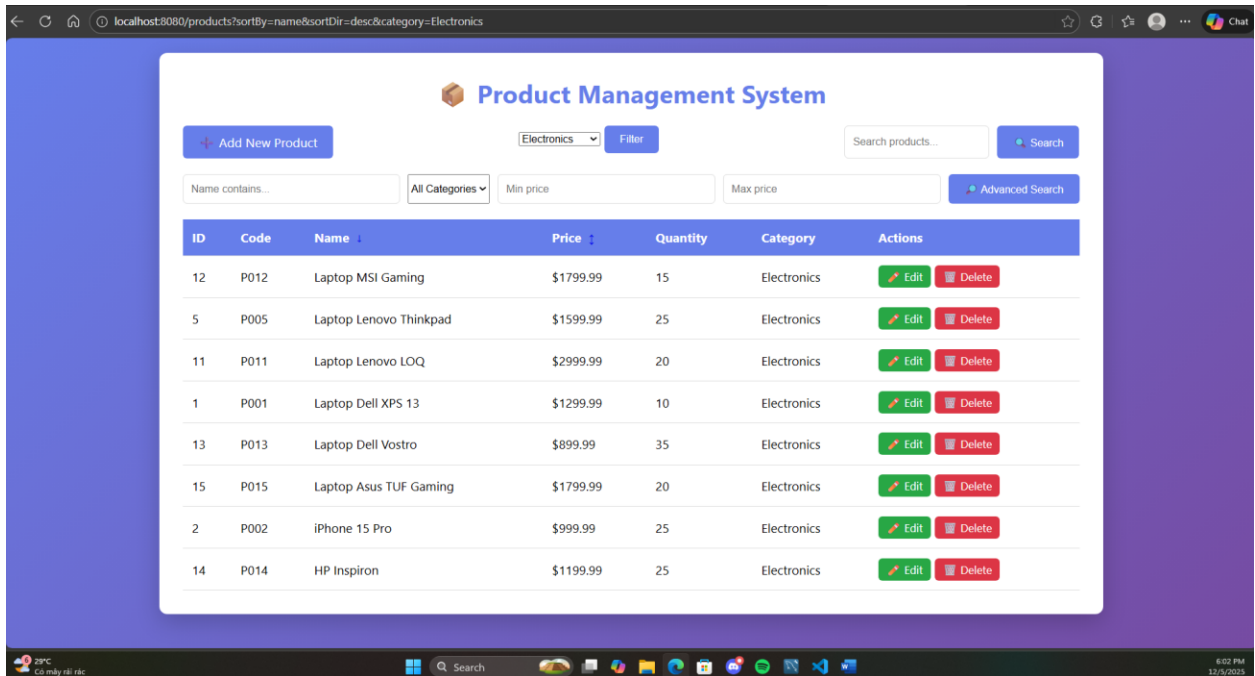
ID	Code	Name	Price	Quantity	Category	Actions
16	P016	Algorithms and Data Structures	\$14.90	35	Books	Edit Delete
9	P009	Egg fried rice	\$2.99	15	Food	Edit Delete
8	P008	Hamburger	\$5.99	15	Food	Edit Delete
14	P014	HP Inspiron	\$1199.99	25	Electronics	Edit Delete
2	P002	iPhone 15 Pro	\$999.99	25	Electronics	Edit Delete
15	P015	Laptop Asus TUF Gaming	\$1799.99	20	Electronics	Edit Delete
13	P013	Laptop Dell Vostro	\$899.99	35	Electronics	Edit Delete
1	P001	Laptop Dell XPS 13	\$1299.99	10	Electronics	Edit Delete
11	P011	Laptop Lenovo LOQ	\$2999.99	20	Electronics	Edit Delete

Product Management System

[Add New Product](#) Electronics Filter Search

All Categories Advanced Search

ID	Code	Name	Price	Quantity	Category	Actions
1	P001	Laptop Dell XPS 13	\$1299.99	10	Electronics	Edit Delete
2	P002	iPhone 15 Pro	\$999.99	25	Electronics	Edit Delete
5	P005	Laptop Lenovo Thinkpad	\$1599.99	25	Electronics	Edit Delete
11	P011	Laptop Lenovo LOQ	\$2999.99	20	Electronics	Edit Delete
12	P012	Laptop MSI Gaming	\$1799.99	15	Electronics	Edit Delete
13	P013	Laptop Dell Vostro	\$899.99	35	Electronics	Edit Delete
14	P014	HP Inspiron	\$1199.99	25	Electronics	Edit Delete
15	P015	Laptop Asus TUF Gaming	\$1799.99	20	Electronics	Edit Delete

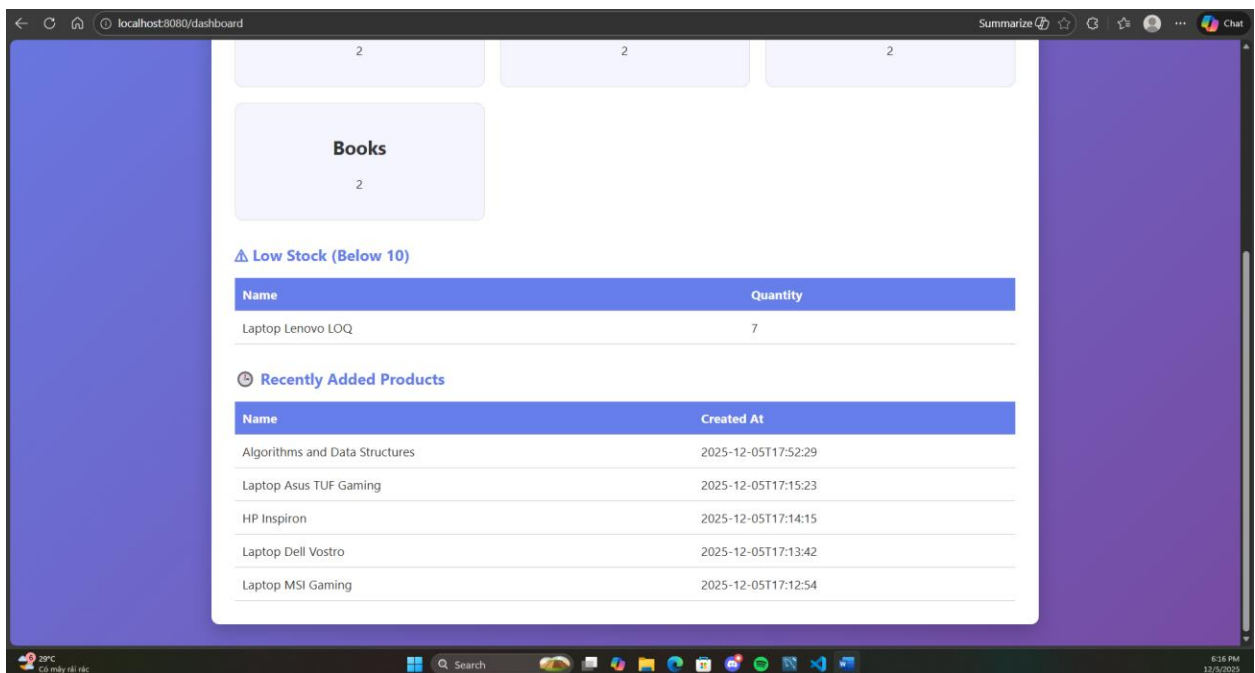
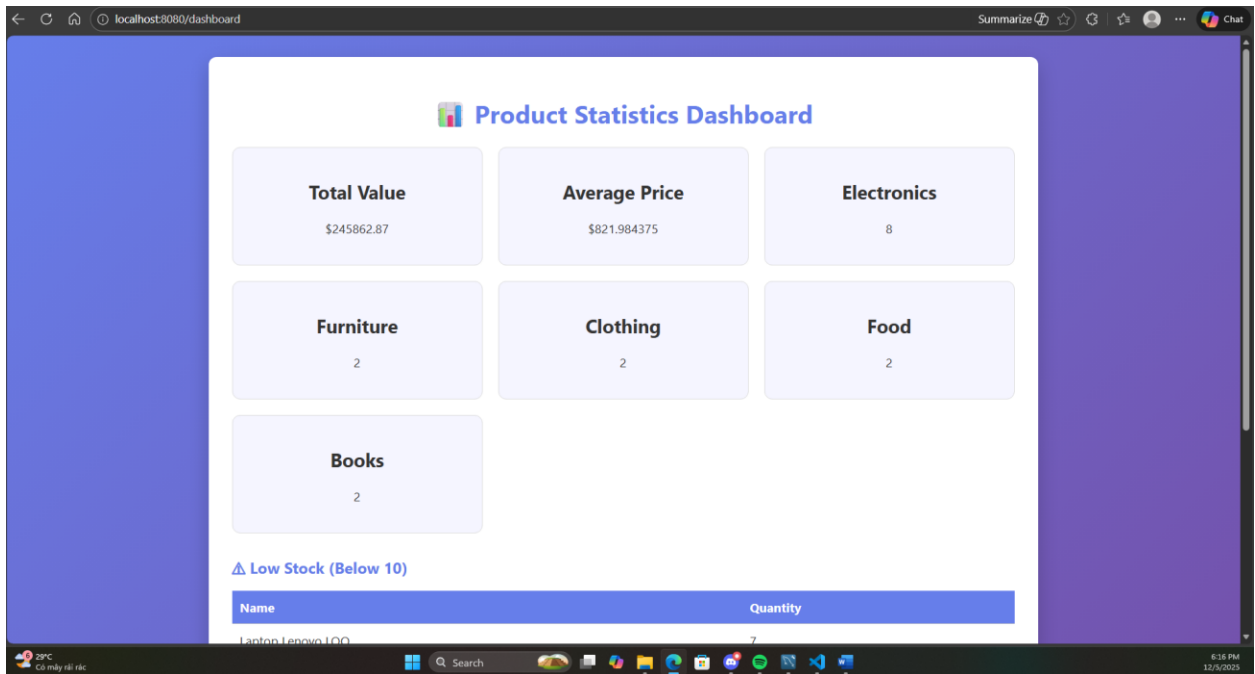


2. Code explanation

- Sorting
 - `Sort sort = Sort.by("field").ascending()`
 - Spring Data handles ordering when fetching list.
- Controller logic
 - If `sortBy` exists, sort is applied.
 - If `category` exists, filtering is applied.
 - Sorting + filtering can run together.
- View links
 - Thymeleaf generates toggling URLs: `sortDir == 'asc' ? 'desc' : 'asc'`

Exercise 8

1. Demo/Test case



2. Code explanation

- Repository queries

- Using @Query for:
 - COUNT
 - SUM
 - AVG
 - Low stock queries

- Example: `SELECT SUM(p.price * p.quantity) FROM Product p`
- Service layer
 - Provides clean methods:
 - `calculateTotalValue();`
 - `calculateAveragePrice();`
 - `findLowStockProducts();`
- Controller
 - Adds all statistics to the model:
`model.addAttribute("totalValue", totalValue);`
- View
 - Displays statistics inside cards and tables.